

A Condensed Theoretical Formulation of the Manifold-Adaptive Weights Empirical Cubature Method (MAW-ECM)

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Abstract

The Manifold-Adaptive Weights Empirical Cubature Method (MAW-ECM) is a hyper-reduction strategy in which the cubature weights are allowed to depend on the position on a nonlinear reduced manifold, while the reduced integration support remains fixed. The method extends the classical Empirical Cubature Method (ECM) by replacing a single cubature vector with a manifold-dependent family of weights,

$$\mathbf{w} = \mathbf{w}(\mathbf{q}_M),$$

where $\mathbf{q}_M$ denotes the latent coordinates parameterizing the reduced manifold.

At the conceptual level, MAW-ECM is formulated as a manifold-adaptive sparsification problem. One seeks the smallest common support capable of satisfying, at every sampled manifold state, a family of linear equality constraints associated with reduced integrals. In addition, a regularization term is introduced to penalize lack of smoothness of the adaptive weight fields over the manifold.

The resulting optimization problem combines combinatorial sparsity, nonnegativity constraints, manifold regularity, and local equality constraints in a single coupled formulation. Rather than solving this problem directly, the implemented methodology adopts a feasible pruning strategy. Starting from an already feasible ECM rule, integration points are removed progressively, while the remaining weights are redistributed so as to preserve feasibility of the constraints.

The present document focuses on the mathematical structure underlying this construction: the original sparsification problem, its intractability, the transition to pruning, the constrained redistribution problem, and the role of regularization during positivity enforcement. Implementation-specific aspects, graph-construction details, regression procedures, and boundary variants are intentionally omitted in order to emphasize the general optimization and hyperreduction structure of the method.

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1 Introduction

The starting point of the present formulation is not the classical Empirical Cubature Method (ECM) itself, but the ECM associated with a nonlinear manifold hyperreduced-order model (HROM).

Consider a nonlinear manifold approximation of the form

$$\mathbf{d}_L \approx \mathcal{D}(\mathbf{q}_M),$$

where

$$\mathbf{q}_M \in \mathbf{R}^{N_{\text{lat}}}$$

denotes the latent coordinates parameterizing the reduced manifold. In a manifold HROM, the reduced internal-force contributions, output integrands, or other nonlinear quantities depend on the manifold position. Consequently, the integrand space itself varies over the sampled manifold.

Let

$$\mathbf{A}_f^{(j)}$$

denote the local integrand data associated with the sampled manifold state

$$\mathbf{q}_M^{(j)}.$$

Collecting all sampled states gives the global manifold integrand matrix

$$\mathbf{A}_{\text{glob}} = [\mathbf{A}_f^{(1)}, \dots, \mathbf{A}_f^{(N_m)}].$$

A standard manifold ECM construction first determines a global reduced basis

$$\mathbf{Q}$$

spanning the relevant manifold integrand content. The fixed ECM rule is then constructed so as to reproduce exactly the reduced integrals associated with this global basis:

$$\mathbf{Q}^T \mathbf{w}^\Phi = \mathbf{Q}^T \mathbf{w}^{\text{FE}}.$$

Here:

- $\mathbf{w}^{\text{FE}} \in \mathbf{R}^{n_{\text{gp}}}$ denotes the finite element integration weights;
- $\mathbf{w}^\Phi \in \mathbf{R}^{n_{\text{gp}}}$ denotes the fixed ECM weights;
- the support of $\mathbf{w}^\Phi$ defines the reduced integration set

$$\mathcal{Z}_{\text{ini}} \subset \mathcal{Z}.$$

At the conceptual level, the corresponding manifold ECM problem may be written as

$$\begin{aligned} \min_{\mathbf{w}^\Phi} \quad & \|\mathbf{w}^\Phi\|_0 \\ \text{subject to} \quad & \mathbf{Q}^T \mathbf{w}^\Phi = \mathbf{Q}^T \mathbf{w}^{\text{FE}}, \\ & \mathbf{w}^\Phi \geq 0. \end{aligned}$$

Thus, the manifold ECM rule is already a compressed representation of the global manifold integrand space.

However, the resulting cubature rule remains fixed over the entire nonlinear manifold. Both the support and the weights are shared globally over all sampled states. This global character may become restrictive. A fixed cubature rule must remain simultaneously accurate for all local manifold configurations and therefore may require substantially more integration points than are locally necessary near individual manifold locations.

The central idea of the Manifold-Adaptive Weights Empirical Cubature Method (MAW-ECM) is to introduce adaptivity at the level of the cubature weights while preserving a common reduced support.

Instead of a single cubature vector, MAW-ECM introduces a manifold-dependent family of weights:

$$\mathbf{w} = \mathbf{w}(\mathbf{q}_M).$$

The integration points remain fixed, but their weights vary over the manifold.

Accordingly, MAW-ECM occupies an intermediate position between:

- a fully global manifold ECM rule, where both support and weights are fixed;
- and a fully local cubature strategy, where both support and weights would be recomputed independently at each manifold state.

The method therefore preserves the notion of a common reduced integration support while allowing local manifold-dependent redistribution of the cubature weights.

The origin of this formulation is closely connected to the Subspace Adaptive Weights ECM (SAW-ECM), in which several reduced subspaces share a common sparse support while possessing distinct cubature weights. In the present setting, the discrete subspace index is replaced by a sampled nonlinear manifold. For each sampled manifold state $\mathbf{q}_M^{(j)}$, one introduces a local cubature vector

$$\mathbf{w}^{(j)} \in \mathbf{R}^{n_{\text{gp}}}.$$

At the conceptual level, the manifold-adaptive sparsification problem becomes

$$\begin{aligned} \min_{\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(N_m)}} \quad & \left\| \sum_{j=1}^{N_m} \mathbf{w}^{(j)} \right\|_0 \\ \text{subject to} \quad & \mathbf{U}^{(j)T} \mathbf{w}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_m, \\ & \mathbf{w}^{(j)} \geq 0, \quad j = 1, \dots, N_m. \end{aligned}$$

Here:

- $\mathbf{U}^{(j)}$ contains the local manifold-dependent reduced integrands associated with the sampled state $\mathbf{q}_M^{(j)}$;
- $\mathbf{b}^{(j)}$ contains the corresponding reduced integrals induced by the initial feasible manifold ECM rule.

The support of

$$\sum_{j=1}^{N_m} \mathbf{w}^{(j)}$$

defines the integration points used somewhere on the manifold. Since the weights are nonnegative, this support coincides with the union of the supports of the local adaptive rules.

However, the adaptive manifold setting introduces an additional difficulty. The vectors

$$\mathbf{w}^{(j)}$$

cannot vary arbitrarily from one manifold state to another. Abrupt variations of the adaptive weights over neighboring manifold states may produce irregular or unstable online behavior.

To moderate this effect, a regularization term penalizing lack of smoothness of the adaptive weight fields is introduced. The resulting conceptual formulation reads

$$\begin{aligned} \min_{\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(N_m)}} \quad & \left\| \sum_{j=1}^{N_m} \mathbf{w}^{(j)} \right\|_0 + \frac{\alpha}{2} \mathcal{R}(\mathbf{W}) \\ \text{subject to} \quad & \mathbf{U}^{(j)T} \mathbf{w}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_m, \\ & \mathbf{w}^{(j)} \geq 0, \quad j = 1, \dots, N_m. \end{aligned}$$

Here

$$\mathcal{R}(\mathbf{W})$$

penalizes lack of smoothness of the adaptive weight fields over the manifold.

Problem (1) represents the original conceptual formulation of MAW-ECM. It combines:

- sparsification,
- positivity,
- manifold regularity,
- and local linear equality constraints.

Nevertheless, this formulation is not solved directly in practice. The main difficulty is combinatorial. The zero-norm objective encodes the discrete decision of which integration points remain active over the manifold.

Instead of solving the sparsification problem directly, the methodology adopted here follows a feasible pruning strategy inspired by ECM and subsequent continuous cubature developments. Starting from an already feasible manifold ECM rule, integration points are progressively removed while the remaining weights are redistributed so as to preserve feasibility of the local linear equality constraints.

The remainder of this document develops this pruning formulation from first principles. Particular emphasis is placed on:

- the transition from the original sparsification problem to the pruning strategy;
- the least-change redistribution problem;
- positivity enforcement through active sets;
- and the role of regularization during constrained redistribution.

2 From sparsification to feasible pruning

The formulation in (1) is useful because it identifies the ideal object sought by MAW-ECM: a common sparse support, nonnegative manifold-dependent weights, satisfaction of the local linear equality constraints, and a regularity mechanism for the weight fields. However, it is not the formulation that is solved in practice.

The main difficulty is the support-selection term

$$\left\| \sum_{j=1}^{N_m} \mathbf{w}^{(j)} \right\|_0.$$

This term encodes a discrete combinatorial decision: which integration points should remain active over the manifold. Even before considering regularization, the direct minimization of this support-union cardinality is not a practical route for constructing a cubature rule.

MAW-ECM therefore replaces the direct sparsification problem by a feasible pruning process. The distinction is important:

sparsification defines the objective, pruning defines the algorithmic route.

The method starts from a feasible cubature rule and removes points only when the remaining weights can be redistributed while preserving feasibility.

2.1 Initialization from a feasible ECM rule

Let

$$\mathcal{Z}_{\text{ini}} \subset \mathcal{Z}$$

be the support of an initial fixed ECM rule, with weights

$$\mathbf{w}_{\text{ini}} \in \mathbf{R}^{|\mathcal{Z}_{\text{ini}}|}.$$

This initial rule is assumed to satisfy the reduced integration constraints used to construct it, together with positivity:

$$\mathbf{w}_{\text{ini}} \geq 0.$$

The adaptive construction is initialized by assigning the same weights to all sampled manifold states:

$$\mathbf{w}^{(j)} = \mathbf{w}_{\text{ini}}, \quad j = 1, \dots, N_{\text{m}}.$$

Equivalently,

$$\mathbf{W} = [\mathbf{w}_{\text{ini}}, \dots, \mathbf{w}_{\text{ini}}] \in \mathbf{R}^{|\mathcal{Z}_{\text{ini}}| \times N_{\text{m}}}.$$

Thus, at the beginning of the pruning process, MAW-ECM coincides with the fixed-weight ECM rule. Adaptivity is introduced only after integration points are removed and the remaining weights are allowed to separate from their common initial value.

2.2 Restriction to the candidate support

All subsequent unknowns are defined only on the initial ECM support. Hence, if

$$N_{\text{cand}} = |\mathcal{Z}_{\text{ini}}|,$$

then

$$\mathbf{w}^{(j)} \in \mathbf{R}^{N_{\text{cand}}}, \quad j = 1, \dots, N_{\text{m}}.$$

The full finite element integration grid is no longer part of the optimization variables. It enters only through the construction of the local linear equality constraints.

For each manifold state $\mathbf{q}_M^{(j)}$, the corresponding constraint system is written as

$$\mathbf{U}^{(j)T} \mathbf{w}^{(j)} = \mathbf{b}^{(j)}.$$

Here $\mathbf{U}^{(j)}$ has already been restricted to the candidate support $\mathcal{Z}_{\text{ini}}$. In the present condensed formulation, we assume that these local constraint systems are constructed using the projected option only. Thus, the right-hand side is defined consistently with the current candidate rule rather than by introducing an additional full-grid target at this stage.

The role of (2.2) is to enforce reproduction of the selected reduced integrals at each sampled manifold state. In optimization terms, these are linear equality constraints.

2.3 One-point pruning trial

At a generic pruning stage, let

$$\mathcal{I}_{\text{act}} \subset \{1, \dots, N_{\text{cand}}\}$$

denote the currently active candidate indices. A pruning trial selects one candidate point

$$p \in \mathcal{I}_{\text{act}}$$

and attempts to remove it. The updated active set would be

$$\mathcal{I}_{\text{act}}^+ = \mathcal{I}_{\text{act}} \setminus \{p\}.$$

Equivalently, the pruning condition is

$$W_{pj}^+ = 0, \quad j = 1, \dots, N_{\text{m}}.$$

The remaining weights are then recomputed so that

$$\mathbf{U}^{(j)T} \mathbf{w}^{(j),+} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_m,$$

and, when positivity is enforced,

$$\mathbf{w}^{(j),+} \geq 0, \quad j = 1, \dots, N_m.$$

The candidate is accepted only if such a redistribution exists and satisfies the prescribed admissibility criteria. Otherwise, the candidate is rejected and another candidate may be tested.

2.4 Candidate ranking

A simple global indicator is used to rank candidate points before testing them. For each currently active point i , define its aggregate weight over the sampled manifold by

$$R_i = \sum_{j=1}^{N_m} W_{ij}.$$

Candidates are tested in increasing order of R_i . Thus, the algorithm first attempts to remove points that carry the smallest total contribution over the manifold.

This ranking is only a heuristic ordering. A small aggregate weight does not guarantee removability. A point with small total weight may still be essential for satisfying the linear equality constraints at some manifold states. Therefore, the final decision is always based on feasibility of the redistribution problem, not on the ranking criterion alone.

2.5 Candidate sweeping

If the first candidate in the ordered list cannot be removed, the algorithm does not terminate immediately. Instead, the next candidate is tested. This candidate sweeping is a practical way of exploiting the combinatorial nature of the sparsification problem without solving the full combinatorial problem.

At a given pruning stage, the algorithm performs:

$$p_1, p_2, \dots$$

where the candidates are ordered by increasing aggregate weight. For each candidate p_k , a redistribution problem is solved. If the redistribution is feasible, the removal is accepted and the pruning stage is completed. If no candidate in the list admits a feasible redistribution, the pruning process stops.

Thus, the method does not assume that the smallest-weight point is always the correct point to remove. It only uses the aggregate weight as a guide for searching the finite set of possible one-point eliminations.

2.6 Stopping criteria

The pruning process terminates in either of two situations.

First, it may stop because no feasible elimination is found. This is the natural feasibility-driven stopping criterion:

$$\text{no candidate } p \in \mathcal{I}_{\text{act}} \text{ admits an admissible redistribution.}$$

Second, a prescribed target number of retained points may be imposed:

$$|\mathcal{I}_{\text{act}}| \leq n_{\text{target}}.$$

When this target is active, the algorithm stops even if further feasible eliminations might exist.

The first mechanism reflects the intrinsic limit imposed by the constraints. The second mechanism is user-prescribed and may be used to stop the pruning at a desired level of compression.

2.7 Interpretation

The pruning formulation preserves the structure of the original sparsification problem while avoiding its direct solution. Each accepted pruning step reduces the common support by one point:

$$\mathcal{Z}_{\text{ini}} \supset \mathcal{Z}_1 \supset \mathcal{Z}_2 \supset \cdots \supset \mathcal{Z}.$$

At the same time, feasibility of the linear equality constraints is maintained through redistribution of the remaining weights.

The result is a sequence of constrained feasible rules with progressively smaller common support. The adaptive character of MAW–ECM appears because the columns of $\mathbf{W}$ are allowed to differ from one another after pruning:

$$\mathbf{w}^{(j)} \neq \mathbf{w}^{(k)} \quad \text{in general.}$$

This is the mechanism by which MAW–ECM trades spatial complexity for parametric complexity: fewer integration points are retained, but their weights become functions over the manifold.

3 Local linear equality constraints

The pruning algorithm operates on local linear equality constraints defined at each sampled manifold state. These constraints are not constructed directly from the original nonlinear integrands. Instead, they are induced by an already feasible manifold ECM rule.

Let

$$\mathbf{A}_f^{(j)} \in \mathbf{R}^{n_{\text{gp}} \times m_j}$$

denote the local integrand data associated with the sampled manifold state $\mathbf{q}_M^{(j)}$. Collecting all sampled states gives the global manifold integrand matrix

$$\mathbf{A}_{\text{glob}} = [\mathbf{A}_f^{(1)}, \dots, \mathbf{A}_f^{(N_m)}].$$

The initial manifold ECM rule is constructed from a reduced basis

$$\mathbf{Q}$$

obtained from the global manifold integrand content, for instance through orthonormalization or truncated compression:

$$\mathbf{Q} = \text{orth}(\mathbf{A}_{\text{glob}}).$$

The corresponding fixed ECM rule satisfies

$$\mathbf{Q}^T \mathbf{w}^\Phi = \mathbf{Q}^T \mathbf{w}^{\text{FE}}.$$

Let

$$\mathcal{Z}_{\text{ini}}$$

and

$$\mathbf{w}_{\text{ini}}$$

denote the resulting support and weights.

The purpose of MAW–ECM is not to reconstruct the original full integration problem again. Rather, it seeks an additional manifold-adaptive compression of this already feasible manifold ECM rule.

3.1 Scaffold constraints

A central ingredient of the construction is the introduction of a scaffold of constraints that must remain exactly satisfied independently of the local manifold integrand content.

The most fundamental scaffold constraint is the constant function:

$$\mathbf{1}.$$

Its inclusion guarantees preservation of the total measure represented by the cubature rule. In particular, it prevents the degenerate zero-weight solution and preserves the global normalization of the adaptive weights.

More generally, additional scaffold constraints may be incorporated whenever certain integration properties are considered essential independently of the local manifold variations.

For instance:

- in elastoplastic problems, one may wish to preserve exact integration of linear stress or internal-variable modes;
- in damage problems, one may enforce exact integration of selected low-order dissipative fields;
- in history-dependent constitutive settings, one may preserve exact integration of specific slowly-varying components regarded as structurally important.

The role of the scaffold is therefore to enforce a minimal integration structure that remains protected throughout the pruning process. The manifold-dependent integrand content is then added on top of this scaffold.

Denote by

$$\mathbf{U}_s$$

the scaffold matrix. At minimum,

$$\mathbf{U}_s = \mathbf{1}.$$

In more general situations,

$$\mathbf{U}_s = [\mathbf{1}, \mathbf{u}_{s1}, \mathbf{u}_{s2}, \dots],$$

where the additional columns correspond to prescribed integration modes that must remain exactly reproduced.

3.2 Local manifold-dependent constraints

For each manifold sample, the projected local integrand content is defined as

$$\widehat{\mathbf{A}}_f^{(j)} := \mathbf{Q}\mathbf{Q}^T \mathbf{A}_f^{(j)}.$$

The local constraint basis is then constructed by augmenting the scaffold with the local projected manifold content:

$$\mathbf{U}^{(j)} = \text{orth} \left[\mathbf{U}_s, \widehat{\mathbf{A}}_f^{(j)} \right].$$

Thus, the local constraints contain:

- a fixed protected scaffold shared globally over the manifold;
- and a manifold-dependent contribution associated with the local reduced integrands.

Restricting to the candidate support inherited from the initial ECM rule gives

$$\mathbf{U}^{(j)} = \mathbf{U}^{(j)}(\mathcal{Z}_{\text{ini}}, :).$$

3.3 Local right-hand sides

The local right-hand side is induced by the initial feasible manifold ECM rule:

$$\mathbf{b}^{(j)} = \mathbf{U}^{(j)T} \mathbf{w}_{\text{ini}}.$$

Thus, the adaptive weights at manifold sample $\mathbf{q}_M^{(j)}$ are required to satisfy

$$\mathbf{U}^{(j)T} \mathbf{w}^{(j)} = \mathbf{b}^{(j)}.$$

By construction, the initial manifold ECM weights satisfy these constraints exactly:

$$\mathbf{U}^{(j)T} \mathbf{w}_{\text{ini}} = \mathbf{b}^{(j)}.$$

The pruning process therefore asks whether the same protected scaffold and local projected manifold integration conditions can still be satisfied after progressively removing integration points from the common support.

This viewpoint is important conceptually. MAW-ECM does not attempt to preserve arbitrary local integrands exactly. Instead, it preserves:

- (i) a protected scaffold of structurally important integration modes;
- (ii) together with the projected manifold-dependent integrand content induced by the initial feasible manifold ECM rule.

4 Redistribution after a tentative pruning step

Assume that, at a given pruning stage, the current active support is

$$\mathcal{I}_{\text{act}} \subset \{1, \dots, N_{\text{cand}}\}.$$

For each sampled manifold state $\mathbf{q}_M^{(j)}$, the current adaptive weights are denoted by

$$\mathbf{w}_{\text{old}}^{(j)} \in \mathbf{R}^{|\mathcal{I}_{\text{act}}|}.$$

They satisfy the local linear equality constraints

$$\mathbf{U}_{\mathcal{I}_{\text{act}}}^{(j)T} \mathbf{w}_{\text{old}}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_{\text{m}},$$

where $\mathbf{U}_{\mathcal{I}_{\text{act}}}^{(j)}$ denotes the restriction of $\mathbf{U}^{(j)}$ to the active support.

A pruning trial selects one candidate point

$$p \in \mathcal{I}_{\text{act}}$$

and attempts to remove it. The remaining active set is

$$\mathcal{I}_{\text{rem}} = \mathcal{I}_{\text{act}} \setminus \{p\}.$$

The question is whether one can construct new weights

$$\mathbf{w}_{\text{new}}^{(j)} \in \mathbf{R}^{|\mathcal{I}_{\text{rem}}|}$$

such that

$$\mathbf{U}_{\mathcal{I}_{\text{rem}}}^{(j)T} \mathbf{w}_{\text{new}}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_{\text{m}}.$$

If such weights exist and satisfy the required admissibility conditions, the candidate removal is accepted.

4.1 Equality-only least-change redistribution

The simplest redistribution problem ignores positivity and regularization. At each manifold state, one seeks the new weights that satisfy the linear equality constraints while remaining as close as possible to the previous weights restricted to the retained support:

$$\mathbf{w}_{\text{old}\mathcal{I}_{\text{rem}}}^{(j)} := \mathbf{w}_{\text{old}}^{(j)}(\mathcal{I}_{\text{rem}}).$$

The local problem is

$$\begin{aligned} \min_{\mathbf{w}_{\text{new}}^{(j)}} \quad & \frac{1}{2} \left\| \mathbf{w}_{\text{new}}^{(j)} - \mathbf{w}_{\text{old}\mathcal{I}_{\text{rem}}}^{(j)} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{U}_{\mathcal{I}_{\text{rem}}}^{(j)T} \mathbf{w}_{\text{new}}^{(j)} = \mathbf{b}^{(j)}. \end{aligned}$$

This is an equality-constrained projection onto an affine subspace.

Let

$$(\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)}) := \mathbf{U}_{\mathcal{I}_{\text{rem}}}^{(j)T}.$$

Then the problem becomes

$$\begin{aligned} \min_{\mathbf{w}_{\text{new}}^{(j)}} \quad & \frac{1}{2} \left\| \mathbf{w}_{\text{new}}^{(j)} - \mathbf{w}_{\text{old}\mathcal{I}_{\text{rem}}}^{(j)} \right\|_2^2 \\ \text{subject to} \quad & (\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)}) \mathbf{w}_{\text{new}}^{(j)} = \mathbf{b}^{(j)}. \end{aligned}$$

Provided that $(\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)})$ has full row rank, the solution is

$$\mathbf{w}_{\text{new}}^{(j)} = \mathbf{w}_{\text{old}\mathcal{I}_{\text{rem}}}^{(j)} - (\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)})^T \left((\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)}) (\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)})^T \right)^{-1} \left((\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)}) \mathbf{w}_{\text{old}\mathcal{I}_{\text{rem}}}^{(j)} - \mathbf{b}^{(j)} \right).$$

This formula is the orthogonal projection of the restricted old weights onto the affine space defined by the local linear equality constraints.

4.2 Minimum-norm particular solution

For the regularized formulation introduced later, it is useful to separate a particular feasible component from the homogeneous admissible variations.

Consider the local linear system

$$(\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)}) \mathbf{w}^{(j)} = \mathbf{b}^{(j)}.$$

If $(\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)})$ has full row rank, the minimum-norm particular solution is

$$\wp^{(j)} = (\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)})^T \left((\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)}) (\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)})^T \right)^{-1} \mathbf{b}^{(j)}.$$

This is the unique solution of minimum Euclidean norm among all vectors satisfying the local linear equality constraints.

Let

$$\mathbf{N}^{(j)}$$

be a basis for the nullspace of $(\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)})$:

$$(\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)}) \mathbf{N}^{(j)} = \mathbf{0}.$$

Then every feasible local redistribution can be written as

$$\mathbf{w}_{\text{new}}^{(j)} = \wp^{(j)} + \mathbf{N}^{(j)} \mathbf{y}^{(j)}.$$

The vector $\mathbf{y}^{(j)}$ contains the remaining free variables after the linear equality constraints have been enforced.

This representation will be used in the regularized active-set formulation. It allows the equality constraints to be satisfied exactly by construction, while the remaining degrees of freedom are selected by minimizing the regularized objective.

4.3 Feasibility check

A necessary condition for feasibility at manifold state j is

$$|\mathcal{I}_{\text{rem}}| \geq r_j,$$

where r_j is the number of independent linear equality constraints at that state. More precisely, the restricted constraint matrix must satisfy

$$\text{rank}(\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)}) = r_j.$$

If this rank condition fails, the candidate point cannot be removed, because the remaining support does not contain enough independent degrees of freedom to satisfy the local constraints.

Thus, each pruning trial involves a local feasibility test at every sampled manifold state. The candidate is admissible only if all local systems remain feasible.

4.4 Role of the equality-only stage

The equality-only redistribution is inexpensive and often sufficient during the early stages of pruning. It does not enforce positivity, but it identifies whether the linear equality constraints can still be satisfied after removing a candidate point.

If the resulting weights remain nonnegative, the pruning step may be accepted without further correction. If negative weights appear, explicit positivity enforcement is required. This leads to the regularized active-set formulation developed in the next section.

The equality-only stage therefore acts as the first and cheapest feasibility test. It removes points as long as admissible redistributions can be obtained without activating inequality constraints.

5 Regularized redistribution with positivity enforcement

The equality-only redistribution introduced in the previous section does not enforce positivity of the adaptive weights. During the early pruning stages, this is often sufficient: the redistributed weights remain nonnegative and the candidate removal can be accepted immediately. However, as the support becomes smaller, the equality-only redistribution may leave the positive orthant and produce negative weights.

When this occurs, positivity enforcement becomes necessary.

The appearance of negative weights should not be interpreted as a numerical failure. It is a structural consequence of attempting to satisfy the same local linear equality constraints with fewer active integration points. The remaining weights are forced to compensate for the removed support, and the unconstrained least-change redistribution may therefore become incompatible with nonnegativity.

The purpose of the present section is to describe the regularized constrained redistribution used once positivity enforcement becomes active.

5.1 Constrained redistribution problem

Assume that a candidate point

$$p \in \mathcal{I}_{\text{act}}$$

has been tentatively removed. The remaining active support is

$$\mathcal{I}_{\text{rem}} = \mathcal{I}_{\text{act}} \setminus \{p\}.$$

The new adaptive weights must satisfy the local linear equality constraints

$$\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)} \mathbf{w}_{\text{new}}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_{\text{m}},$$

together with positivity:

$$\mathbf{w}_{\text{new}}^{(j)} \geq 0, \quad j = 1, \dots, N_m.$$

In addition, one seeks redistributions that remain close to the previous weights while avoiding excessively irregular variations over the manifold. Accordingly, the constrained redistribution problem is written as

$$\begin{aligned} \min_{\mathbf{W}} \quad & \frac{1}{2} \sum_{j=1}^{N_m} \left\| \mathbf{w}_{\text{new}}^{(j)} - (\mathbf{w}_{\text{old}})_{\mathcal{I}_{\text{rem}}}^{(j)} \right\|_2^2 + \frac{\alpha}{2} \mathcal{R}(\mathbf{W}) \\ \text{subject to} \quad & \mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)} \mathbf{w}_{\text{new}}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_m, \\ & \mathbf{w}_{\text{new}}^{(j)} \geq 0, \quad j = 1, \dots, N_m. \end{aligned}$$

The first term is a least-change objective. It minimizes the perturbation introduced by the pruning step. The second term penalizes lack of smoothness of the adaptive weight fields over the manifold.

5.2 Regularization term

The regularization functional

$$\mathcal{R}(\mathbf{W})$$

penalizes rough variations of the adaptive weights over neighboring manifold states. Its precise algebraic form is not essential for the present condensed formulation. The important point is that it couples nearby manifold samples and discourages abrupt variations of the redistributed weights.

Without regularization, positivity enforcement may produce strongly irregular adaptive fields. This occurs because active-set transitions are local events: a weight may become clamped to zero at one manifold state while remaining free at neighboring states. As pruning progresses, these local transitions may generate oscillatory or discontinuous patterns in the adaptive weights.

The regularization term moderates this effect by favoring redistributions that remain smooth over the sampled manifold while still satisfying the linear equality and positivity constraints.

5.3 Nullspace parametrization

The linear equality constraints can be eliminated exactly through a nullspace parametrization.

For each manifold state, let

$$\wp^{(j)}$$

be the minimum-norm particular solution satisfying

$$\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)} \wp^{(j)} = \mathbf{b}^{(j)}.$$

Let

$$\mathbf{N}^{(j)}$$

be a basis for the nullspace of the local constraint matrix:

$$\mathbf{A}_{\mathcal{I}_{\text{rem}}}^{(j)} \mathbf{N}^{(j)} = 0.$$

Then every feasible redistribution can be written as

$$\mathbf{w}_{\text{new}}^{(j)} = \wp^{(j)} + \mathbf{N}^{(j)} \mathbf{y}^{(j)}.$$

The constrained redistribution problem is therefore rewritten in terms of the reduced variables

$$\mathbf{y}^{(j)}.$$

The linear equality constraints are automatically satisfied, and only positivity and regularization remain active in the optimization problem.

This parametrization is important computationally because it separates:

$$\text{exact feasibility} \quad \text{from} \quad \text{regularized admissible redistribution.}$$

5.4 Active-set enforcement

Positivity is enforced through an active-set strategy.

Suppose that, after redistribution, some component becomes negative:

$$W_{ij} < 0.$$

That degree of freedom is then declared active:

$$W_{ij} = 0,$$

and removed from the free variables of the constrained redistribution problem. The redistribution is solved again on the remaining admissible unknowns.

This process is repeated until either:

- all weights are nonnegative,
- or no feasible redistribution exists.

It is important to distinguish between:

- pruning of an integration point,
- and local active-set clamping.

Pruning removes an integration point globally:

$$W_{pj} = 0, \quad j = 1, \dots, N_m.$$

The point disappears completely from the common support.

By contrast, active-set enforcement is local in the manifold direction:

$$W_{ij} = 0$$

may hold only at selected manifold samples. The integration point remains part of the common support, even though its weight vanishes locally at some manifold states.

This distinction is essential. The support-selection mechanism and the positivity-enforcement mechanism are different operations.

5.5 Acceptance or rejection of the pruning step

After positivity enforcement and regularized redistribution, the candidate removal is accepted only if:

- (i) all local linear equality constraints remain satisfied;
- (ii) all redistributed weights are nonnegative;
- (iii) the resulting redistribution is admissible according to the chosen regularization strategy.

If these conditions are satisfied, the pruning step is committed:

$$\mathcal{I}_{\text{act}} \leftarrow \mathcal{I}_{\text{rem}}.$$

Otherwise, the candidate is rejected and another candidate point is tested.

Thus, every accepted pruning step produces a feasible adaptive cubature rule. The support shrinks monotonically, while the remaining weights adapt progressively over the manifold.

5.6 Interpretation

The regularized redistribution problem is the central mechanism of MAW–ECM. It is the stage where the method departs fundamentally from a fixed-weight ECM construction.

A fixed ECM rule compresses globally:

$$\text{one support,} \quad \text{one weight vector.}$$

By contrast, MAW–ECM introduces manifold-dependent redistributions:

$$\text{one support,} \quad \text{many locally adapted weight vectors.}$$

The purpose of the regularization is not to impose additional physics or to modify the reduced integrals themselves. Its role is purely structural: among all feasible positive redistributions, it favors those producing smoother adaptive weight fields over the manifold.

In this way, MAW–ECM trades fixed global integration accuracy for adaptive local redistribution while preserving a common reduced integration support.

6 Pruning algorithm

The previous sections describe the two basic ingredients of MAW–ECM:

- a family of local linear equality constraints defined over the sampled manifold;
- a constrained redistribution problem used after a tentative point removal.

This section combines these ingredients into the complete pruning algorithm.

The algorithm starts from an initial feasible ECM rule

$$(\mathcal{Z}_{\text{ini}}, \mathbf{w}_{\text{ini}}),$$

replicates its weights over the sampled manifold, and progressively eliminates integration points. Each accepted elimination reduces the common support by one point. The adaptive weights are recomputed after each tentative removal.

6.1 Main loop

At the beginning,

$$\mathcal{I}_{\text{act}} = \{1, \dots, N_{\text{cand}}\}, \quad \mathcal{I}_{\text{elim}} = \emptyset,$$

where

$$N_{\text{cand}} = |\mathcal{Z}_{\text{ini}}|.$$

The adaptive weights are initialized as

$$\mathbf{w}^{(j)} = \mathbf{w}_{\text{ini}}, \quad j = 1, \dots, N_{\text{m}}.$$

At each pruning iteration, the aggregate weight of each active candidate is computed:

$$R_i = \sum_{j=1}^{N_{\text{m}}} W_{ij}.$$

The active candidates are sorted in increasing order of R_i . The algorithm then tests candidate removals in this order.

For a candidate p , the point is tentatively removed:

$$\mathcal{I}_{\text{rem}} = \mathcal{I}_{\text{act}} \setminus \{p\}.$$

A redistribution problem is solved on the remaining support. If the redistribution satisfies the local linear equality constraints, positivity, and the regularization-based admissibility requirements, the removal is accepted:

$$\mathcal{I}_{\text{act}} \leftarrow \mathcal{I}_{\text{rem}}, \quad \mathcal{I}_{\text{elim}} \leftarrow \mathcal{I}_{\text{elim}} \cup \{p\}.$$

If not, the candidate is rejected and the next candidate in the ordered list is tested.

The pruning process stops when either:

- the prescribed target number of points has been reached;
- or no candidate admits a feasible redistribution.

6.2 Algorithmic statement

Algorithm 1 MAW-ECM feasible pruning algorithm

Require: Initial support $\mathcal{Z}_{\text{ini}}$; initial weights $\mathbf{w}_{\text{ini}}$; sampled manifold states $\{\mathbf{q}_M^{(j)}\}_{j=1}^{N_m}$; local constraint systems $\{(\mathbf{U}^{(j)}, \mathbf{b}^{(j)})\}_{j=1}^{N_m}$; target number of points n_{target} .

Ensure: Reduced support $\mathcal{Z}$; adaptive weights $\mathbf{W}$ on $\mathcal{Z}$.

1: Initialize

$$\mathcal{I}_{\text{act}} \leftarrow \{1, \dots, |\mathcal{Z}_{\text{ini}}|\}, \quad \mathcal{I}_{\text{elim}} \leftarrow \emptyset.$$

2: Initialize adaptive weights

$$\mathbf{w}^{(j)} \leftarrow \mathbf{w}_{\text{ini}}, \quad j = 1, \dots, N_m.$$

3: **while** $|\mathcal{I}_{\text{act}}| > n_{\text{target}}$ **do**

4: Compute aggregate weights

$$R_i = \sum_{j=1}^{N_m} W_{ij}, \quad i \in \mathcal{I}_{\text{act}}.$$

5: Sort active candidates in increasing order of R_i .

6: Set accepted $\leftarrow$ false.

7: **for** each candidate p in the ordered list **do**

8: Define

$$\mathcal{I}_{\text{rem}} \leftarrow \mathcal{I}_{\text{act}} \setminus \{p\}.$$

9: Solve the equality-only redistribution problem on $\mathcal{I}_{\text{rem}}$.

10: **if** the resulting weights satisfy positivity **then**

11: Accept the candidate:

$$\mathcal{I}_{\text{act}} \leftarrow \mathcal{I}_{\text{rem}}.$$

12: Update $\mathbf{W}$.

13: Set accepted $\leftarrow$ true.

14: **break**

15: **else**

16: Solve the regularized active-set redistribution problem on $\mathcal{I}_{\text{rem}}$.

17: **if** a feasible positive redistribution is found **then**

18: Accept the candidate:

$$\mathcal{I}_{\text{act}} \leftarrow \mathcal{I}_{\text{rem}}.$$

19: Update $\mathbf{W}$.

20: Set accepted $\leftarrow$ true.

21: **break**

22: **end if**

23: **end if**

24: **end for**

25: **if** accepted = false **then**

26: Stop pruning.

27: **end if**

28: **end while**

29: Set

$$\mathcal{Z} \leftarrow \mathcal{Z}_{\text{ini}}(\mathcal{I}_{\text{act}}).$$

30: Keep the final adaptive weights restricted to $\mathcal{I}_{\text{act}}$.

6.3 Interpretation

The algorithm is greedy in the sense that it accepts one feasible point removal at a time. However, it is not purely based on the aggregate weight ranking. The ranking only provides an ordering of candidates. The actual decision is made by solving the redistribution problem.

Thus, pruning is governed by feasibility:

a point is removed only if the remaining support can still carry admissible weights.

This is the main difference between the original sparsification statement and the practical MAW–ECM algorithm. The former seeks the optimal sparse support in one step; the latter constructs a decreasing sequence of feasible supports.

The candidate sweeping mechanism is essential. If the first candidate fails, the algorithm tests alternatives rather than stopping immediately. This reflects the combinatorial nature of the support-selection problem: failure of one candidate does not imply that no point can be removed.

6.4 Output of the pruning stage

At termination, the algorithm returns:

$$\mathcal{Z} = \mathcal{Z}_{\text{ini}}(\mathcal{I}_{\text{act}}),$$

and a matrix of adaptive weights

$$\mathbf{W}_{\text{red}} = \mathbf{W}(\mathcal{I}_{\text{act}}, :).$$

Each column of $\mathbf{W}_{\text{red}}$ corresponds to one sampled manifold state, and each row corresponds to one retained integration point.

The resulting cubature rule has the form

$$I(\mathbf{q}_M^{(j)}) \approx \sum_{i \in \mathcal{Z}} W_{ij} f_i(\mathbf{q}_M^{(j)}), \quad j = 1, \dots, N_m.$$

The support is common to all sampled states, but the weights are state-dependent.

This is the final object constructed by the condensed MAW–ECM formulation.

7 Interpretation and conceptual position of MAW–ECM

The previous sections describe the mathematical structure and the constructive algorithm underlying MAW–ECM. It is useful, however, to clarify the precise conceptual position occupied by the method within the broader landscape of hyperreduction and reduced integration techniques.

7.1 Global versus local cubature

Classical ECM constructs a single sparse cubature rule:

$$(\mathcal{Z}, \mathbf{w}^\Phi).$$

Both the support and the weights are fixed. The rule must therefore remain simultaneously accurate over the entire reduced manifold explored during the offline stage.

This global character is both the strength and the limitation of fixed-weight ECM. It provides a unique reduced integration rule, but the resulting support may remain substantially larger than what is locally necessary at individual reduced states.

At the opposite extreme, one could imagine a fully local strategy in which the cubature rule is recomputed independently at every manifold point:

$$(\mathcal{Z}(\mathbf{q}_M), \mathbf{w}(\mathbf{q}_M)).$$

In that setting, both the support and the weights become state-dependent. Such a formulation would potentially yield extremely small local cubature rules, but at the price of losing a common integration structure altogether.

MAW-ECM occupies an intermediate position between these two extremes:

$$\text{fixed support,} \quad \text{adaptive weights.}$$

The support

$$\mathcal{Z}$$

is shared globally over the manifold, whereas the weights vary with the latent coordinates.

Thus, the method preserves a common reduced integration structure while allowing local adaptation of the integration action.

7.2 Compression mechanism

The essential compression mechanism of MAW-ECM is different from that of standard ECM.

In fixed-weight ECM, compression is entirely spatial:

$$\text{fewer integration points} \implies \text{same weights everywhere.}$$

The rule must contain enough points to reproduce accurately all relevant integrands over the sampled manifold simultaneously.

By contrast, MAW-ECM introduces an additional degree of freedom:

$$\text{the weights themselves vary over the manifold.}$$

The support no longer needs to reproduce all local integrand configurations through a single fixed set of coefficients. Instead, the integration action is adapted locally through manifold-dependent redistributions.

In this sense, MAW-ECM trades:

$$\text{spatial complexity} \quad \text{for} \quad \text{parametric complexity.}$$

Fewer integration points are retained, but the associated weights become fields defined over the latent manifold.

7.3 Role of regularization

The regularization term does not modify the local linear equality constraints. Its role is purely structural.

Without regularization, positivity enforcement may produce highly irregular adaptive weights. The origin of these irregularities is the active-set nature of the constrained redistribution problem. As pruning progresses, some weights become clamped to zero at isolated manifold locations while remaining free at neighboring states.

Consequently, the adaptive weight fields may exhibit abrupt local transitions:

$$W_{ij} = 0 \quad \text{at one manifold sample,}$$

while

$$W_{ik} > 0$$

at nearby samples.

The regularization term penalizes such rough redistributions and favors smoother adaptive fields. It therefore acts as a stabilizing mechanism for the manifold-dependent cubature rule.

Importantly, the regularization does not define the feasible set. Feasibility is determined exclusively by:

- the linear equality constraints,
- positivity,
- and the active support.

The regularization only selects, among feasible redistributions, those that remain smoother over the manifold.

7.4 Pruning versus active-set enforcement

Two different sparsity mechanisms coexist in MAW-ECM and should not be confused.

The first mechanism is pruning:

$$W_{pj} = 0, \quad j = 1, \dots, N_m.$$

This removes an integration point globally from the common support. Once accepted, the point disappears from the reduced cubature rule.

The second mechanism is local active-set enforcement:

$$W_{ij} = 0$$

at selected manifold samples only. In this case, the integration point remains part of the common support, even though its weight vanishes locally at some states.

Thus:

pruning changes the support,

whereas

active sets modify local admissible redistributions.

This distinction is essential for understanding the geometry of the adaptive weight fields. The final support remains fixed globally, but the local activity patterns of the weights may still vary over the manifold.

7.5 Interpretation as a manifold-adaptive cubature rule

The final output of MAW-ECM is a cubature rule of the form

$$\int_{\Omega} f(x, \mathbf{q}_M) dx \approx \sum_{i \in \mathcal{Z}} w_i(\mathbf{q}_M) f(x_i, \mathbf{q}_M).$$

The integration points are fixed, whereas the weights depend on the latent coordinates.

This dependence is the essential novelty introduced by the method. The reduced cubature rule is no longer a static object. It becomes a manifold-adaptive integration operator.

At the same time, MAW-ECM remains fundamentally different from a black-box regression of weights. The adaptive fields are not introduced heuristically. They are generated through constrained feasible redistributions preserving the local reduced integration structure inherited from the initial ECM rule.

Thus, the method should be interpreted as:

adaptive hyperreduction under constrained cubature structure,

rather than as a purely data-driven interpolation of quadrature coefficients.

7.6 Final perspective

The central observation motivating MAW–ECM is that the number of integration points required by a fixed cubature rule does not necessarily scale with the intrinsic dimension of the reduced manifold.

A fixed rule must remain valid globally:

$$\text{one support,} \quad \text{one weight vector,} \quad \text{all manifold states.}$$

This global requirement may force the support to represent the union of many locally distinct integrand configurations.

MAW–ECM relaxes this restriction by allowing the weights to adapt locally while preserving a common support. The method therefore attempts to reduce the gap between:

$$\text{the intrinsic manifold dimension}$$

and

$$\text{the number of retained integration points.}$$

The resulting formulation combines:

- sparse reduced integration,
- manifold adaptivity,
- constrained redistribution,
- positivity preservation,
- and regularized feasible pruning.

This completes the condensed theoretical formulation of the Manifold-Adaptive Weights Empirical Cubature Method.

8 A physical interpretation of the pruning process

Before continuing with the mathematical details of the redistribution and active-set procedures, it is useful to pause and develop a more physical and perhaps more appealing interpretation of the MAW–ECM mechanism.

The goal of the present section is not mathematical rigor. Rather, the idea is to help the reader build an intuitive mental picture of what the algorithm is actually doing. In practice, many aspects of the method become much easier to understand once one stops thinking in terms of matrices and optimization variables and instead imagines a physical structure evolving under multiple loading conditions.

The analogy that follows should therefore be understood mainly as a conceptual device intended to facilitate understanding of the method without entering immediately into all the subtleties of constrained optimization.

8.1 A suspended bridge under many wind conditions

Imagine a very large suspended bridge composed of a dense network of tensile cables. Suppose now that the bridge must remain in equilibrium not under one single loading condition, but under an entire family of nearby conditions: slightly different wind directions, slightly different traffic distributions, slightly different deformation states, and so on.

This is the role played by the manifold samples

$$\mathbf{q}_M^{(j)}.$$

Each manifold point may be interpreted as one loading configuration of the same bridge.

Now imagine that, initially, the bridge has been designed in a very conservative way. There are many more cables than are truly necessary. The structure is highly redundant.

However, not all cables play the same role.

Some cables are absolutely crucial. They carry large portions of the load and participate in balancing many equilibrium conditions simultaneously. Others are less important and contribute only marginally.

This is precisely what one observes in the initial ECM rule.

Empirically, the ECM weights are highly nonuniform: some weights become much larger than others. This is not accidental. It is a direct consequence of the greedy nature of ECM. The algorithm progressively selects those integration points that are most effective for reproducing the reduced equilibrium conditions.

Under the structural analogy, the ECM algorithm has already identified the principal load paths of the bridge.

The resulting structure is therefore not homogeneous. Some cables carry large tensions, while others play only secondary corrective roles.

8.2 The frozen-force philosophy of standard ECM

At this point, one may imagine the philosophy of a standard fixed-weight ECM rule in the following way.

Suppose the engineer designing the bridge imposes the following artificial restriction:

every cable must always carry exactly the same tension pattern for all loading configurations.

Immediately, the bridge becomes much harder to optimize.

Why?

Because different loading conditions naturally require different stress redistributions. A cable that is extremely important under one wind direction may become almost irrelevant under another.

Nevertheless, the fixed-weight ECM philosophy forces the same force distribution to remain valid everywhere on the manifold.

As a consequence, the bridge requires many more cables than are locally necessary under individual loading states.

This is precisely the main limitation of a fixed manifold ECM rule.

8.3 The central idea of MAW-ECM

The central idea of MAW-ECM is almost deceptively simple.

Instead of forcing the cable tensions to remain frozen over all loading conditions, one allows the internal force distribution to adapt to the loading configuration.

The bridge keeps the same surviving cables:

same support,

but the tensions carried by those cables may vary from one loading state to another:

adaptive weights.

Thus, the adaptive weights

$$\mathbf{w}^{(j)}$$

may be interpreted as the tensions carried by the surviving cables under the loading configuration associated with manifold point $\mathbf{q}_M^{(j)}$.

This immediately introduces additional flexibility into the system.

A cable that is critically important for one loading condition may carry very little tension under another. The structure can now reorganize its internal forces locally over the manifold.

8.4 Pruning as cutting cables

Now imagine beginning a process of progressive simplification.

One cable is selected and cut.

Immediately, the external loading conditions remain exactly the same. The wind does not disappear merely because one cable has been removed. Therefore, the remaining cables must reorganize their tensions so as to preserve equilibrium.

At the beginning, this is easy.

The bridge is highly redundant. There are many alternative equilibrium paths. When one cable disappears, neighboring cables simply increase their tensions slightly and the structure remains perfectly stable.

This is the meaning of the null space in the MAW–ECM formulation.

The null-space directions represent admissible internal force redistributions that preserve the same external equilibrium resultants. The structure can “wobble” its internal stresses while keeping the same global balance.

However, every removed cable reduces the redundancy of the structure.

The remaining bridge becomes progressively more stressed:

- fewer cables remain;
- each cable carries more responsibility;
- the redistribution freedom decreases;
- the null space shrinks.

Eventually, the bridge approaches a critical regime in which the surviving cables are almost uniquely determined by equilibrium.

8.5 Slacking cables and active sets

Now comes the most important part of the analogy.

Suppose that, after removing another cable, the redistribution problem attempts to assign a negative tension to one of the surviving members.

Mechanically, this means:

the structure is asking a cable to push.

But a cable cannot push.

A cable can only pull.

This is precisely the meaning of a negative weight in MAW–ECM.

The positivity constraint

$$W_{ij} \geq 0$$

therefore expresses a tension-only admissibility condition.

The active-set mechanism then acquires a completely natural interpretation.

Whenever a cable attempts to enter compression, it simply goes slack:

$$W_{ij} = 0.$$

Importantly, this does not necessarily remove the cable globally from the bridge. The same cable may:

- carry tension under some loading configurations,
- become slack under others,
- and reactivate elsewhere on the manifold.

This distinction is fundamental.

Pruning corresponds to permanently cutting a cable from the bridge.

The active set, by contrast, determines which surviving cables are actually load-bearing under each local loading configuration.

8.6 Regularization as coherent structural behavior

Finally, the role of regularization also becomes very intuitive.

Neighboring manifold points correspond to nearby loading configurations. One therefore expects nearby loading conditions to produce nearby internal stress patterns.

Without regularization, however, tiny changes in the loading state may produce wildly different redistributions:

- one cable active here,
- slack nearby,
- active again elsewhere.

The structure would behave erratically.

The regularization discourages this behavior. It favors coherent stress evolution throughout the manifold and promotes smoother transitions between neighboring loading configurations.

8.7 Final picture

Under this interpretation, MAW–ECM may be viewed as a progressive search for the smallest admissible tensile skeleton capable of supporting an entire family of neighboring equilibrium states.

The initial ECM rule provides an already compressed bridge obtained through a greedy equilibrium process. The pruning stage progressively cuts cables from the structure. The remaining cables continuously reorganize their tensions so as to preserve equilibrium. The null space measures the remaining redundancy of the bridge. The active set describes locally slack cables. The regularization promotes coherent structural behavior over the manifold.

Although purely conceptual, this picture captures surprisingly well the true mechanism underlying the MAW–ECM redistribution process.